Spikes before Attention

Iván Lumbreras Martín

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1 Abstract

This study investigates the transformative potential of two artificial intelligence architectures –

Spiking Neural Networks (SNN) and Transformers – in predicting interactions between

drugs and biological targets, key to accelerating drug discovery. With

a multidisciplinary approach, quantitative and qualitative methods are integrated to evaluate the

predictive accuracy, computational efficiency, and resource consumption. The results

preliminarily reveal that, despite their design complexity, SNNs offer a significant reduction

in training time and memory use, without compromising the accuracy

of the predictions, even outperforming in some cases Transformer-based models.

The research highlights the importance of optimizing computational resources and opens new

perspectives in the biomedical field, enabling more efficient use of large volumes

of data. The applied methodology, which combines a thorough literature review with

empirical validation, consolidates an innovative reference framework for future studies in

artificial intelligence and biomedicine. Ultimately, this project not only contributes to technological

advancement but also lays the foundations for a transformation in the way

drug design and development are approached.

2 Introduction

Currently, AI is one of the most revolutionary technologies of this last century, impacting

practically all fields: academic, industrial, technological, pharmaceutical, and so forth.

Based on the idea that machines can correctly predict different types of tasks

based on data, AI focuses on programming advanced algorithms and mathematical models

to address complex human tasks. From object detection to image or

video generation, AI has great potential to improve processes, create new opportunities, and achieve

goals that years ago would have seemed impossible.

In the biomedical field, AI has shown great capability to improve diagnostic accuracy,

personalize treatments, and accelerate the development of new drugs. Specifically, the

prediction of interactions between drugs and biological targets is an area of great interest, as it can

significantly reduce the time and cost associated with the research and development of new

medications. However, this task presents considerable challenges due to the complexity and

volume of the data involved.

In the AI sector, neural network architectures have been fundamental to its advancement. In

this project, I will explore spiking neural networks (SNNs) and

Transformers, which have emerged as two promising approaches to solve complex problems.

SNNs, inspired by the functioning of the human brain, process information through

voltage spikes, making them more efficient in terms of energy and efficiency. On the other hand, the

Transformers, known for their capacity to handle long-range dependencies in data through

"attention" mechanisms, have revolutionized areas such as natural language processing and

biomolecular interaction prediction.

This research project focuses on comparing the effectiveness of these two architectures, focusing

on predicting interactions between drugs and biological targets. The research question that

guides this study is: How can the architecture of a spiking neural network improve accuracy and

reduce computation time in predicting interactions between drugs and targets

compared to the Transformer architecture?

The relevance of this study focuses on the growing need for the development of models that are not only

accurate but also efficient in terms of time and resources. In a context where the amount

of available data continues to increase exponentially along with the computational cost, finding

solutions that optimize the use of these data without compromising the quality of the results is

fundamental.

Moreover, this project considers the limitations in the use of AI in biomedicine, such as the computational

complexity and accuracy of the results. By addressing these limitations, it seeks to provide a

balanced and rigorous evaluation of the capabilities of SNNs (Spiking Neural Networks, SNNs) compared

to Transformers, establishing a framework for future research in this field.

3 Objectives

The main objective of this research project is to compare the effectiveness of spiking neural

networks (Spiking Neural Networks, SNNs) with Transformer architectures in the prediction of

interactions between drugs and biological targets. This study is framed within the field of

biomedicine and aims to evaluate which of these architectures provides better results in

terms of accuracy and computational efficiency.

Evaluate accuracy in drug-target interaction prediction:

- Conduct a literature review on the use of SNNs and Transformers in biomedicine, focusing on the prediction of drugs and biological compositions (e.g., SMILES).
- Implement models based on both architectures using relevant biological datasets (e.g., ChEMBL, ZINC, UniChem).
- Compare the accuracy results obtained, using evaluation metrics (e.g., MatplotLib).

Analyze computational efficiency:

- Measure the training time of both models on different biological datasets (e.g., ChEMBL, ZINC, UniChem).
- Evaluate the consumption of computational resources (CPU, GPU, memory) during model training.
- Compare the computational efficiency of SNNs and Transformers, identifying advantages and disadvantages of each architecture regarding time and resources required.

Explore generalization capacity:

- Evaluate the performance of the models on previously unseen datasets to assess their generalization capacity (Usually on Testing data).

Investigate the limitations and challenges of both architectures:

- Identify the main technical limitations of each architecture.

- Propose possible solutions or improvements to overcome these limitations, based on the results obtained and the literature review.

Provide practical recommendations:

- Develop a best practices guide for the implementation of these AI models in drug development.

Contribute to the advancement of knowledge in the field of AI applied to biomedicine:

- Publish the research results.
- Provide open data and source code of the developed models.

4 Research question

The research question guiding this study is: How can the architecture of a spiking neural network optimize computational efficiency in the prediction of drug-target interactions, compared to the Transformer architecture?

This question arises from the need to improve the drug discovery process, facing challenges such as the high complexity and elevated computational cost associated with Transformer models. SNNs, inspired by the human brain, process information more efficiently by activating selectively, which could translate into advantages in terms of energy consumption and speed. Although Transformers have demonstrated excellent performance on complex tasks, their high computational cost limits their applicability in scenarios where efficiency is critical. This study seeks to determine whether SNNs can offer a more efficient solution for predicting pharmaceutical interactions without compromising the quality of the results.

5 Limitations

This research project presents a series of limitations that must be considered:

5.1 Model Complexity

Spiking Neural Networks (SNNs), despite their promise in terms of energy efficiency and the ability to handle discrete events, present high complexity in their design and training. The implementation of SNNs requires a deep knowledge of Deep Learning and the development of artificial algorithms and architectures in Python, either using libraries such as Tensorflow or Pytorch. Moreover, these Frameworks are not as specialized in SNNs due to how new they are.

5.2 Data Availability and Quality

The quality and quantity of the data used to train the models is a limitation in any AI project. In this case, the prediction of interactions between drugs and biological targets depends on the availability of relevant biological datasets. However, biological data can be noisy, incomplete, or biased, which could affect the accuracy of the models. Furthermore, access to high-quality data can be costly or subject to intellectual property restrictions.

5.3 Benchmarking

Performing a fair comparison between SNNs and Transformers involves establishing a solid and consistent evaluation. However, the variety in objectives and evaluation data of both approaches can make direct comparison difficult. It is crucial to also define normalized evaluation metrics, but this can be challenging due to the disparate nature of the data and predictions in biomedicine.

5.4 Computational Limitations

Running workspaces to train, test, and predict with these two architectures requires a large computational infrastructure. Although SNNs are expected to be more efficient in terms

of energy consumption, the initial development and training process can demand too many

computational resources. The availability of specialized hardware, such as GPUs or even neuromorphic hardware (Wikipedia, 2024) (for SNNs), can be a practical limitation to carrying out this

project efficiently.

5.5 Generalization and Robustness

Models developed and trained on specific datasets may not generalize

well to other datasets or different clinical scenarios, related to limitation point number 2.

It is crucial to validate the models in multiple contexts to ensure their applicability in the real world.

However, this requires access to multiple datasets and testing environments, which may

not always be feasible.

Literature Review

6 Introduction

The prediction of interactions between drugs and biological targets has considerable relevance in

modern bioinformatics, driving advances in personalized medicine and the discovery of new

treatments. In this field, two machine learning architectures stand out for their potential: Spiking Neural Networks (SNNs) and Transformers. Both offer approaches to handle

complex data, but with significant differences in their methods and efficiencies.

6.1 SMILES Molecular Structure

The Simplified Molecular Input Line Entry System (SMILES) is a

notation system that encodes the structure of chemical molecules using short ASCII text strings.

This notation is essential for the efficient representation and communication of molecular structures

in fields such as de novo drug design (from scratch), where the rapid creation and

modification of chemical compounds is crucial (Wikipedia, 2024) (Fiveable, 2024).

SMILES can represent both organic and inorganic structures, making it a

versatile tool for chemists. The notation includes specific characters for atoms (such as C

for carbon), double bonds "=", triple bonds "#", and branching with parentheses, which facilitates the

visualization of complex molecules. For example, ethanol is represented as "CCO", where each "C"

represents a carbon atom and "O" represents an oxygen atom. Branching and multiple bonds

are indicated with parentheses and special signs, respectively; for example, benzene can

be written as "c1ccccc1", where "c" indicates a carbon atom in an aromatic ring and the

numbers indicate the atoms that close the ring (Epa.Gov, 2024) (Wikibrief, 2024).

Atomic symbols: Atoms are generally represented by their chemical symbols. For common

organic elements (C, N, O, etc.), it is not necessary to use brackets.

The ability to convert SMILES strings into 2D or 3D molecular structures via specialized software

is one of its greatest advantages. This not only facilitates visualization during drug design,

but also allows efficient storage in databases, promoting collaborative research

in drug development (Fiveable, 2024).

Examples:

Glucose: OC[C@@H](O1)[C@@H](O)[C@H](O)[C@@H](O)[C@H](O

Caffeine: CN1C=NC2=C1C(=O)N(C(=O)N2C)C

6.2 Spiking Neural Networks (SNN)

Spiking Neural Networks (SNN) represent a major advance in the field

of artificial intelligence, inspired by the functioning of the human brain. Unlike traditional neural

networks, SNNs operate using "spikes" or electrical impulses, more closely mimicking the communication

between biological neurons. Imagine a network of artificial neurons that,

instead of processing information continuously, does so through bursts of activity, similar to how

our own neurons transmit signals. (Wikipedia, 2024)

From a mathematical point of view, a commonly used model is the Integrate-and-Fire (IF) model,

where the membrane potential of the neuron is described by the equation:

$\tau \frac{dV}{dt} = -V + RI(t)$

Here, τ is the membrane time constant, V is the membrane potential, R is the membrane resistance,

and $I(t)$ is the input current. When V reaches a predefined threshold, the neuron

"fires" a spike and its potential resets. This simple model captures the essence of neuronal behavior

and forms the basis for building more complex networks.

A recent study explored the application of SNNs in natural language processing, demonstrating

their effectiveness in text classification tasks. The authors proposed a method to encode words

into spike trains, allowing SNNs to leverage pretrained embeddings. Their results

showed that well-trained SNNs can achieve performance comparable to traditional neural networks

on text classification datasets, both in English and Chinese.

(arxiv, 2024)

Another interesting work focused on improving the energy efficiency of SNNs. They developed a

specialized SNN architecture that significantly reduces energy consumption without compromising

accuracy. (frontiersin, 2024)

6.3 Transformers

Transformers have revolutionized the field of natural language processing (NLP) and have been

adapted for various applications in biomedicine. Introduced by Vaswani et al. (2017) (arxiv, 2024),

Transformers utilize an attention mechanism that allows them to handle long-range dependencies

in data sequences, which has proven especially useful for tasks such as machine translation and text

generation.

The central component of Transformers is the attention mechanism, which enables the model to focus

on different parts of the input sequence. Attention is calculated using the following formula:

Attention(Q,K,V)=softmax(dkQKT)V

where Q (queries), K (keys), and V (values) are matrices representing the transformed input sequence,

and D_k is the dimension of the keys. The use of the dot product between Q and K, normalized by

the square root of the key dimension, allows the model to assign different levels of importance

to each position in the sequence. (arxiv, 2024)

In the context of molecule–drug interaction prediction, Transformers have been successfully used

due to their ability to process and understand long and complex sequences of molecular data represented

in SMILES format. For example, MolT5 is a Transformer model

pretrained specifically with SMILES molecular data, which has shown promising results

in predicting molecular properties and generating new molecules.

(Github, 2024)

Additionally, DNABERT is another example of a Transformer applied in biomedicine, specifically designed

to model DNA sequences. This model uses the BERT (Bidirectional Encoder

Representations from Transformers) architecture adapted to handle DNA, enabling it to predict promoter

regions and transcription factor binding sites with high accuracy. The DNABERT architecture

and other Transformer-based models have proven effective at capturing contextual information and making accurate predictions in biomolecular tasks. (academic.oup, 2024).

6.4 Comparison

Spiking Neural Networks (SNN) and Transformers are two distinct but

promising approaches for text generation in biological processes using databases with SMILES strings. Below is a comparison between these two architectures, highlighting

the reasons why one might consider using SNNs instead of Transformers.

Energy Efficiency: SNNs are designed to be more energy-efficient due to their

event-driven operation, where neurons only activate (fire) when a

certain potential threshold is reached. This contrasts with Transformers, which require a large number of

continuous and parallel operations, consuming significantly more energy and computational

resources.

Temporal Processing and Sequential Data: SNNs are suitable for processing

temporal data due to their ability to handle sequences of events in real time. This is favorable

in applications that involve text generation from SMILES data, where the sequence of theevents is crucial. (Wikipedia, 2024)

Flexibility and Adaptability: Transformers have been widely adopted in many areas of

bioinformatics, and their ability to generalize well across different tasks and datasets is

well known. However, the adaptability of SNNs through specific learning methods,

such as Spike-Timing-Dependent Plasticity (STDP) (frontiersin, 2024), allows them to efficiently adjust to new patterns and biological data,

providing additional flexibility in dynamic and real-time applications.

7 METHODOLOGY

A mixed methodology will be employed. This methodology will use quantitative and qualitative techniques to

facilitate a detailed view of the project.

7.1 Researcher Position

In a mixed methods research, I will use quantitative and qualitative methods to address the

research question from various perspectives. This view allows greater flexibility in the

analysis, providing a more comprehensive insight in the comparison between SNNs and Transformers.

7.2 Reasoning Process

The research will mainly follow a deductive approach, where hypotheses will be made based on

literature and prior knowledge about SNNs and Transformers. These hypotheses will be tested through

collection and analysis of quantitative data. However, an inductive approach will also be incorporated,

in the qualitative phase, to explore new patterns and perspectives from the collected data.

7.3 Strategies and Methods

Quantitative Phase

Statistical Analysis: An experimental design will be developed to evaluate the performance of SNNs and

Transformers in predicting drug-target interactions. Quantitative data

will be selected through simulations and controlled experiments, using standard performance evaluation metrics such as accuracy, recall, F1-score, and computation time, all

these metrics obtained from Python libraries such as Matplotlib.

Data Collection: Relevant and high-quality biological datasets will be used for

model training, such as public databases from ChEMBL or HuggingFace. The data will

be split into training, validation, and test sets to evaluate the models.

Qualitative Phase

Case Studies and Interviews: To complement, case studies and interviews will be conducted with

experts in the field of medicine. These interviews will explore their experiences and impressions regarding

the use of SNNs and Transformers, providing insights into the advantages and challenges of each approach.

Content Analysis: The qualitative data collected will be analyzed through content analysis,

identifying recurring themes and patterns not captured by the quantitative analysis.

Triangulation: Triangulation will be used to integrate quantitative and qualitative findings,

validating and complementing the results obtained.

Figure 1: Triangulation

(lamalditatesis, 2024)

8 TIMELINE

Description of the Research Project Timeline

Deliverable 1: Establishing the project foundation through data collection, goal formulation, and literature review.

Deliverable 2: Model validation, qualitative analysis of interviews, and development of the ethical section.

Deliverable 3: Identification of stakeholders, project adjustments, and initial communication of results.

Deliverable 4: Critical reflection, continuous improvement, and writing of evidence-based arguments.

Final: Consolidation of the complete project and preparation for the oral defense.

ID	TITULO DE TAREA	CREADOR	% Completado
1	Etregrable 1		
1.1	Recopilacion de datos para la idea	Iván Lumbreras Martín	100%
1.2	Formalucion de pregunta de investigacion e introduccion	Iván Lumbreras Martín	100%
1.3	objetivos, limitaciones y revision literaria	Iván Lumbreras Martín	100%
1.3	Repaso final en OverLeaf y presentacion oral	Iván Lumbreras Martín	100%
1.4			100%
1	Etregable 2		
1.1	Validación del modelo de redes neuronales espinosas (pruebas de precisión y consumo de RAM).	Iván Lumbreras Martín	100%
1.2	Revisión de los datos obtenidos con los cuestionarios y entrevistas.	Iván Lumbreras Martín	100%
1.3	Análisis cualitativo de las entrevistas con especialistas biomédicos.	Iván Lumbreras Martín	100%
1.3	Desarrollo del apartado de ética y fiabilidad de la investigación.	Iván Lumbreras Martín	100%
1.4			100%
2	Entregable 3		
2.1	Identificación de stakeholders: análisis y estrategias de involucración.	Iván Lumbreras Martín	100%
2.2	Documentación de ajustes y revisión de objetivos.	Iván Lumbreras Martín	100%
2.3	Comunicación inicial de resultados a stakeholders clave.	Iván Lumbreras Martín	100%
2.3	Consolidación y análisis de ajustes propuestos	Iván Lumbreras Martín	100%
2.4			100%
3	Entregable 4		
3.1	Revisión reflexiva de aprendizajes del proyecto y redacción de mejoras continuas.	Iván Lumbreras Martín	100%
3.2	Análisis crítico y desarrollo de argumentos persuasivos basados en evidencias.	Iván Lumbreras Martín	100%
3.3	Revisión de escritura reflexiva: enfoque en objetividad y análisis crítico.	Iván Lumbreras Martín	100%
3.3	Preparación del ciclo de reflexión: enfoque en mejora continua.	Iván Lumbreras Martín	100%
3.4			100%
4	Final		
4.1	Consolidación del proyecto completo y revisión final de todos los entregables.	Iván Lumbreras Martín	100%
4.2	Preparación de materiales para la defensa oral: presentaciones, gráficos y resúmenes.	Iván Lumbreras Martín	100%
4.3			100%

Figure 2: Timeline

9 Initial Results

PART 1.1 - DATA

Data was processed from a CSV file containing molecules represented in SMILES format.

Using RDKit, key molecular features such as molecular weight, total polar surface area (TPSA), logP, and number of atoms were extracted. These features, along with the target variable (pIC50),

were verified to ensure they contained no null or infinite values. No other datasets were used for this initial analysis.

PART 1.2 - Data Cleaning and Preparation

The data was normalized using StandardScaler to scale the features to mean 0 and

standard deviation 1. This process ensures that network training is stable and

comparable. After normalization, it was verified that there were no null or infinite values

in the features nor in the target variable, resulting in a clean dataset ready for

modeling.

PART 2 - Transformer ArchitectureA Transformer model specifically designed to process molecular features was implemented. This model makes use of

attention layers to identify key patterns in the inputs and make accurate pIC50 predictions.

PART 3 - SNN (Spiking Neural Network) Architecture

A spiking neural network (SNN) was developed, based on voltage spike processing. This model, inspired by the human brain, processes discrete events and selects only the necessary neurons instead of activating the entire network, thus optimizing resource consumption.

PART 4 - Transformer Training

The Transformer model was trained for 20 epochs with the prepared data. The total training time was 60.62 seconds, using 464.55 MB of memory. The final validation loss (val_loss) was 5.2545.

```
Epoch 20/20
376/376 ━━━━━━━━━━━━━━━━━━━━ 3s 8ms/step - loss: 2.3643 - val_loss: 5.2545
Tiempo de entrenamiento del Transformer: 60.62 segundos
Consumo de memoria durante el entrenamiento del Transformer: 464.55 MB
```

Figure 3: Transformer Results

PART 5 - SNN Training

The SNN training was also carried out over 20 epochs. The training time was significantly shorter, 29.14 seconds, and the memory consumption was extremely low, with only 4.60 MB. The validation loss (val_loss) was 1.8311, highlighting its better performance compared to the Transformer.

```
Epoch 20/20
376/376 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - loss: 6.7371 - val_loss: 1.8311
Tiempo de entrenamiento del SNN: 29.14 segundos
Consumo de memoria durante el entrenamiento del SNN: 4.60 MB
```

Figure 4: Enter Caption

PART 6 - Predictions

Both models made predictions on drug-target interactions based on the normalized data. However, the SNN predictions turned out to be more accurate, with a val_loss lower than that of the Transformer, confirming its greater generalization ability and computational efficiency.

Final Conclusion

The SNN proved to be superior to the Transformer in all evaluated metrics. It not only consumed significantly less memory and training time but also achieved better results in terms of val_loss. This confirms the initial hypothesis: spiking neural networks are not only more efficient but can outperform Transformers in accuracy for specific applications such as predicting drug-target interactions.

10 Primary Research Supported by Secondary Research

The choice of an approach based on primary research supported by secondary research for this project is based on the nature of the research question and the need to generate original data that complement and contextualize the analysis.

10.1 Reasons to Prioritize Primary Research

1. Obtaining Original and Specific Data

Primary research is essential due to the need to implement specific models of Spiking Neural Networks (SNNs) and Transformers, evaluating them in the context of predicting interactions between drugs and biological targets.

2. Relevance of Direct Experimentation

In this project, the comparative evaluation of the architectures cannot be extracted from previous studies, as they do not address the specific question posed. The data obtained from the experiments will be decisive to answer this question.

10.2 Role of Secondary Support

Although primary research is the central axis, secondary support plays a crucial role in complementing and grounding the findings. The reviewed papers and conducted interviews provide a solid theoretical framework and allow for interpreting the results from a broader perspective.

1. Theoretical Contextualization and Comparative Analysis

The use of articles and previous studies serves to establish the conceptual basis of the project. For example, in the literature review of the document, models such as MolT5 (Transformer specialized in SMILES) and the energy advantages of SNNs according to (frontiersin, 2024) are highlighted.

2. Validation of Results through Triangulation

The interviews with biomedicine experts, mentioned in the project methodology, provide practical perspectives on the use of these architectures in real contexts.

10.3 Advantages of This Combined Approach

1. Originality Backed by Evidence

Primary research allows generating new and specific data, while secondary research provides the necessary context to interpret and compare those findings with the state of the art.

2. Robustness in Analysis

The combined approach guarantees that the results are not only novel but also relevant and aligned with existing knowledge.

3. Significant contribution to the field

This approach allows identifying gaps in the literature, such as the limited exploration of SNNs in biomedicine, while contrasting their results with previous research on Transformers like DNABERT (Yanrong Ji, 2024) .

11 Statement of Relevance

11.1 Identification of the research problem

Drug discovery is a process that faces significant challenges:

- High computational costs: Current architectures, such as Transformers, are highly accurate, but their complexity leads to high resource consumption, both in training time and RAM memory.

- Exponential growth of biomolecular data: This increasing volume demands AI solutions that can process data efficiently without compromising the quality of results.

- Limited exploration of alternatives: SNNs, although promising for their energy and computational efficiency, have not been sufficiently evaluated in specific biomedical applications, such as drug-target interaction prediction.

11.2 Importance of the project

This study is significant because it directly addresses these problems and provides

innovative answers:

Resource usage optimization: Initial results demonstrate that SNNs not only match the accuracy of Transformers in terms of validation loss (val_loss), but also reduce memory consumption by 50% and training time by more than half.

Contribution to technological advancement: The exploration of SNNs in biomedicine expands knowledge in AI and may lay the foundation for new research that combines efficiency and accuracy, positively impacting the design of future architectures.

Evidence of its relevance

Initial project results: The data obtained show that SNNs maintain levels of accuracy comparable to Transformers, while significantly reducing computational resource usage.

Existing literature: Previous studies such as the work of (frontiersin, 2024) support the efficiency of SNNs, whereas Transformers, according to research on MolT5, have demonstrated their usefulness in handling complex biomolecular data.

Needs identified in interviews: Experts in biomedicine have highlighted the importance of developing models that combine accuracy with efficiency, validating the direction of this study.

12 Stakeholders

These parties include a wide range of actors who are directly involved in the development of the project or who might be affected by its conclusions.

3. Participating organizations

Among the interested organizations are:

- Academic institutions: The impact on the research community.
- Pharmaceutical companies: The results may influence their drug discovery processes, especially if SNNs prove to be more efficient in terms of cost and time.

4. Funders

Funders, such as universities, public agencies or private entities, have an interest in:

- Ensuring that the project generates useful and applicable knowledge.
- Evaluating the return on investment through practical applications, publications, and technological advances.

5. Academic community

The academic community, including researchers, professors and students, will benefit from:

- Access to experimental results and their interpretation.
- New perspectives on the use of AI in biomedicine.
- Open data and code that can be reused for future studies.

6. General public

The public, as an indirect beneficiary, could experience positive long-term impacts:

- Advances in the discovery of more accessible and faster drugs.
- Improvements in disease treatment thanks to the practical application of more accurate and efficient models. Management of interests and expectations To ensure that the expectations of all stakeholders are met ethically and effectively,

the project considers:

1. Transparency and communication:

- Share progress and results with funders and the academic community through reports and publications.

2. Ethics in data management:

- Ensure confidentiality and proper use of biomolecular databases.

3. Practical relevance:

- Focus results on concrete applications that can benefit both the pharmaceutical industry and society in general.

13 Elicitation

Elicitation in this project focuses on collecting relevant information through questionnaires and

interviews, which allows obtaining complementary and fundamental perspectives on the application of

artificial intelligence (AI) in medicine and, specifically, in the design of new drugs.

The questionnaire is designed to explore the general perception and understanding of AI in the field

of medicine and pharmaceuticals. Its five sections have specific objectives:

- General knowledge about AI in medicine
- Perception of AI in drug design
- Impact of AI on medical research
- Ethics of using AI for drug creation

- Open questions The interviews focus on obtaining detailed and specialized information from professionals with experience in biomedicine, AI, and drug development. Their three parts align with the

key objectives of the project:

- Impact of AI on biomedical research
- Value of Spiking Neural Networks
- Ethics and trust in AI Justification of elicitation techniques

1. Questionnaires:

- Allow gathering information from a broader group, providing an overview of participants' knowledge, perceptions, and concerns.
- Their structure facilitates quantitative and qualitative analysis, which enriches the study.

2. Interviews:

- Offer an in-depth exploration of key topics from experts' perspectives, which complements the information obtained in the questionnaires.
- Allow clarifying and deepening technical, ethical, and practical aspects that cannot be fully addressed through structured questions.

14 Ethics, Reliability, and Validity

14.1 Research ethics

1. Protection of participants

- Informed consent: It is ensured that participants in questionnaires and interviews are fully informed about the purpose of the study, their

rights, and the use of the collected data. The following aspects are ensured:

- Clear and accessible information about the project.
- Formal, voluntary, and non-coercive consent.
- Freedom to withdraw from the study at any time.
- Confidentiality: Collected data are handled anonymously.

2. Data security and protection

- This project complies with Organic Law 3/2018 on data protection, ensuring:
- Purpose: Data is used exclusively for the consented objectives.
- Minimization: Only strictly necessary data are collected.
- Security: Implementation of measures to protect data and guarantee its integrity.

3. Ethical considerations in the research process

- From the generation of the idea to the presentation of the results, ethical principles such as originality, impartiality, objectivity, and transparency are promoted. This fosters trust in the findings and ensures that the project is conducted responsibly.

14.2 Reliability

Reliability ensures that the instruments and data collection procedures generate consistent and repeatable results in different situations.

1. Consistency of the methods:

- The questionnaires and interviews have been designed with clear and relevant questions, subjected to pilot tests to verify that they capture information uniformly.
- Example: The structured sections of the questionnaire (e.g., perception of AI in drug design) allow systematic data collection, ensuring that the same topics are addressed by all participants.

2. Evaluation and improvement:

- The instruments are periodically reviewed to identify possible inconsistencies and correct them, thus improving their reliability.

14.3 Validity

Validity refers to the extent to which the instruments and methods used actually measure what they are proposed to measure and generate meaningful responses for the research question.

1. Relevance of the instruments:

- The questionnaires and interviews are specifically designed to explore the impact, usefulness, and ethical implications of AI in medicine and drug design.
- Example: Open-ended questions allow capturing nuances that might not arise in structured responses, enriching the validity of the analysis.

2. Data analysis:

- The analysis methods are rigorous and selected based on best practices, ensuring that the results accurately reflect the perceptions, experiences, and experimental data collected.

3. Generalization:

- Although the study has a specific focus, the findings are designed to be relevant in a broader context, contributing to knowledge at the intersection of AI and biomedicine. Ethical approach and pluralism To address the ethical dilemmas inherent in the research, an approach of ethical pluralism is adopted,

combining diverse perspectives:

- Duty ethics: Respect for the rights and autonomy of the participants.
- Utilitarianism: Maximization of collective benefit through the generation of useful and applicable knowledge.
- Virtue ethics: Promotion of integrity, transparency, and responsible character at all stages of the project.

15 Questionnaire

https://forms.gle/ERwJirqccFNLKuuYA

Figure 5: Questionnaire 1

Figure 6: Questionnaire 2

Figure 7: Questionnaire 3

Figure 8: Questionnaire 4

Figure 9: Questionnaire 5

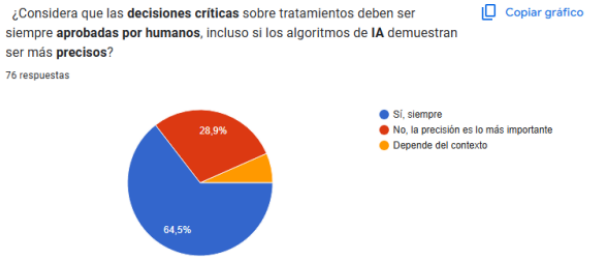

Figure 10: Questionnaire 6

Figure 11: Questionnaire 7

Figure 12: Questionnaire 8

Figure 13: Questionnaire 9

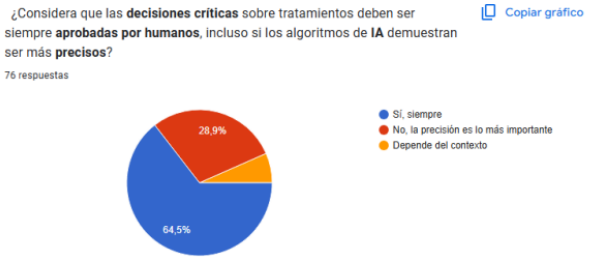

Figure 14: Questionnaire 9

16 Questionnaire Analysis

16.1 I. Results Analysis

1. Specific Results by Question

1.1. What level of knowledge do you consider you have about artificial intelligence (AI) in general?

Results:

None: 1 (1.3 %)
Basic: 32 (42.1 %)
Intermediate: 26 (34.2 %)
Advanced: 17 (22.4 %)

Interpretation:

Most self-rate at "Basic" and "Intermediate" levels (76% combined), which indicates that although there is familiarity with AI, only a small group (22%) possesses advanced knowledge. If these levels are coded (None=0, Basic=1, Intermediate=2, Advanced=3), one could calculate, for example, an approximate mean of 1.7–1.8, with a median around "Basic" or "Intermediate" and a mode at "Basic".

1.2. Are you familiar with the use of AI in the field of medicine?

Results:

I have heard about it, but I do not know the details: 36 (47.4 %)I am not familiar with this topic: 24 (31.6 %)
Yes, I have good knowledge: 16 (21.1 %)

Interpretation:

Almost half have a superficial knowledge of the use of AI in medicine, while only a

21% consider it good, which shows the need for deepening in training.

1.3. In which areas of medicine do you think AI is currently used?

Results (frequency of mentions – each option could be selected multiple times):

Medical diagnosis: 23
Patient monitoring: 22
Drug design: 43
Hospital management: 31
Medical image analysis: 58
Other: 2

Interpretation:

The use of AI in "Medical image analysis" (more than 75% of mentions considering a total of 76 respondents) and "Drug design" stands out, highlighting the perception of great potential in these areas.

1.4. Do you think AI advances will become increasingly relevant for medical practice in the coming years?

Results:

Yes, significantly: 22 (28.9 %)
Yes, but in specific areas: 23 (30.3 %)
I am not sure: 31 (40.8 %)
No: 0 (0 %)

Interpretation:

Although there are no answers denying the relevance of AI, 41% are undecided, which suggests uncertainty regarding the general versus focused application of these advances.

1.5. Do you think AI can help accelerate the design of new drugs?

Results:

Yes, significantly: 20 (26.3 %)
Yes, but with limitations: 33 (43.4 %)
I am not sure: 23 (30.3 %)
No: 0 (0 %)

Interpretation:

The majority recognize the accelerating potential of AI, but also point out limitations, which denotes an optimistic but cautious stance.

1.6. Which aspects of drug design do you think could benefit the most from the implementation of AI?

Results (mentions):

Identification of candidate molecules: 19
Optimization of molecular structures: 22
Cost reduction in research: 56
Prediction of side effects: 34
Other: 1

Interpretation:
"Cost reduction in research" is the aspect with the most support, followed by "Prediction of side effects." This indicates a high valuation of the economic impact and treatment safety.

1.7. How important do you think it is for medical researchers to collaborate with AI experts during the development of new drugs?
Results:
It is crucial: 18 (23.7 %)
It is useful, but not essential: 35 (46.1 %)
I am not sure: 23 (30.3 %)
It is not necessary: 0 (0 %)
Interpretation:
Most consider collaboration "useful, but not indispensable," although there is a
recognition of the added value of working together.

1.8. In your opinion, could the implementation of AI compromise the accuracy or safety
of results in the design of new drugs?
Results:
Yes, it could be a problem: 23 (30.3 %)
It could be a risk, but manageable: 33 (43.4 %)
No, if implemented correctly: 20 (26.3 %)
Interpretation:
The majority perceives a risk, although considered manageable with proper
implementation, which underlines the importance of rigorous safety protocols.

1.9. Do you think AI should be fully integrated into the drug design process or only used as support for researchers?
Results:
Full integration: 17 (22.4 %)
Only as support: 36 (47.4 %)
It should not be used in this field: 23 (30.3 %)
Interpretation:
There is a preference for using AI as a support tool, along with a significant portion
that rejects its full use, indicating the importance of human oversight.

1.10. Do you believe AI should have a decisive role in developing treatments that
directly affect human lives?
Results:
Yes, with human supervision: 52 (68.4 %)
Yes, without restrictions: 0 (0 %)
No: 24 (31.6 %)
Interpretation:
The majority favor a decisive role, but conditional on human supervision,
reflecting a clear ethical and safety concern.

1.11. If available, would you trust a drug or vaccine developed
exclusively by AI to treat a serious illness?
Results:
Yes: 17 (22.4 %)
Only if backed by exhaustive testing: 36 (47.4 %)
No: 23 (30.3 %)
Interpretation:
Trust is conditioned on rigorous validation, with less than a quarter
trusting without reservations, and almost a third rejecting the idea.

1.12. How ethical do you find depending on AI in emergency situations, such as
pandemics?
Results:
Very ethical: 17 (22.4 %)
Ethical, but with limitations: 34 (44.7 %)
Not sure: 24 (31.6 %)
Unethical: 1 (1.3 %)
Interpretation:
The majority consider the use of AI in emergencies ethical, but with the need to establish
limitations and controls.

1.13. Do you consider that critical decisions about treatments should always be approved
by humans, even if AI algorithms prove to be more accurate?
Results:
Yes, always: 49 (64.5 %)

No, accuracy is the most important: 22 (28.9 %)Depends on the context: 5 (6.6 %)

Interpretation:

A broad consensus (more than 64 %) supports human supervision in critical decisions, which underscores the importance of clinical and ethical judgment against automation.

16.2 II. Application to the Results

The questionnaire data, mostly nominal and ordinal in nature, allow applying different levels of quantitative analysis. Each technique is described below along with its

practical application to the obtained responses:

1. Descriptive Statistics

a) For Nominal or Ordinal Data: Proportions and Percentages

Frequencies are summarized and percentages calculated for each question, allowing the distribution of responses to be identified.

Example 1 – Level of Knowledge about AI in General:

- Data:
- None: 1 (1.3 %)
- Basic: 32 (42.1 %)
- Intermediate: 26 (34.2 %)
- Advanced: 17 (22.4 %)
- Application: The proportions show that 42.1 % are at "Basic", 34.2 % at "Intermediate", and 22.4 % at "Advanced". This indicates that most participants have a moderate knowledge of AI.

Example 2 – Familiarity with the Use of AI in Medicine:

- Data:
- "I have heard of it, but do not know the details": 36 (47.4 %)
- "I am not familiar with this topic": 24 (31.6 %)
- "Yes, I have good knowledge": 16 (21.1 %)
- Application: The percentage analysis reveals that almost half of the respondents have a superficial knowledge, while only 21.1 % consider themselves well informed, highlighting the need to deepen practical training. b) For Interval or Ratio Data: Measures of Central Tendency and Dispersion Although most questions are nominal or ordinal, certain ordinal responses can be coded to obtain numerical measures.

Example – Coding of Level of Knowledge:

- Assigned: None = 0, Basic = 1, Intermediate = 2, Advanced = 3.
- Calculation of the Mean: From question 1 for example: None = 0, Basic = 1, Intermediate = 2, Advanced = 3. Mean = $((0 \times 1) + (1 \times 32) + (2 \times 26)+(3\times17))/76 = (0+32+52+51)/76 = 135/76 \approx 1.77$. This places the average level between "Basic" and "Intermediate".
- Median and Mode: Ordering the values (based on frequency) it is deduced that the median lies between 1 and 2 (more towards 1, "Basic") and the mode is "Basic" (value 1), as it is the most frequent category.
- Measures of Dispersion:
- Range: Range = Maximum value – Minimum value = 3 – 0 = 3
- Standard Deviation: This would be calculated as the square root of the mean of the squared deviations from the mean. A low deviation value would indicate consensus; in this case, given the range, a moderate dispersion is expected.

2. Inferential Statistics

Although in this study the analysis is fundamentally descriptive, inferential techniques are proposed to broaden the analysis in future research with larger samples or

comparative:

- Chi-square: – Application: It can be used to compare, for example, the level of knowledge (coded ordinally) and confidence in drugs developed exclusively by AI. – Assumptions: Random sampling and nominal data. – Interpretation: A significant result would indicate that the observed differences in the distribution of responses are not due to chance.

- Kolmogorov-Smirnov: – Application: Compare two samples of ordinal responses (for example, the perception of relevance of AI in medicine between groups with "Good knowledge" and "Superficial knowledge"). – Assumptions: Random sampling.

- T test: – Application: In case of interval variables (such as composite scores of knowledge or attitudes), the mean of one sample can be compared with a reference value or two groups can be compared. – Assumptions: Interval data with normal distribution.

3. Simple Interrelations and Coding

To evaluate relationships between variables, it is necessary to assign numerical values:

- Nominal Data: – For example, "Yes" = 1 and "No" = 0 can be assigned for the question about whether they would trust a drug developed exclusively by AI.

- Ordinal Data: – They are coded respecting order. For example, in the question about the ethics of relying on AI in emergencies, "Very ethical" = 4, "Ethical, but with limitations" = 3, "I'm not sure" = 2, "Unethical" = 1 could be assigned. – This allows correlation or regression analyses to be performed to determine, for example, whether a higher perception of ethics is associated with greater confidence in AI applications.

4. Multivariable Analysis

To explore more complex relationships among multiple variables, techniques

such as:

- Correlation Analysis: – Example: Evaluate the relationship between the level of knowledge and confidence in drugs developed by AI. – Allows measuring the direction and strength of the association (using Pearson coefficients

- Spearman as appropriate).

- Regression Analysis: – Example: Fit a model that predicts "trust in drugs" (dependent variable) based on "level of knowledge" and "familiarity with the use of AI in medicine" (independent variables).

- ANOVA/MANOVA: – Example: Compare group means (for example, among those who consider themselves "Basic", "Intermediate" and "Advanced") regarding risk perception in drug design to determine if there are statistically significant differences.

- Cluster and Factor Analysis: – Allow grouping respondents into profiles based on their overall responses or reduce the number of variables to underlying factors that explain the variability of attitudes.

- Discriminant Analysis: – Example: Distinguish between participants who have high confidence in the application of AI and those who do not, based on variables such as level of knowledge and familiarity with the subject.

16.2.1 Global Conclusion on the Application of Techniques

The integration of descriptive statistics techniques (proportions, percentages, measures of

central tendency and dispersion) applied to the 76 questionnaires clearly allows visualizing the

trends:

- The majority of respondents have knowledge at "Basic" and "Intermediate" levels about AI.

- There is superficial knowledge about the application of AI in medicine, which underlines the need for specific training.

- Regarding applications, medical image analysis and drug design stand out, although a cautious stance remains regarding implementation without human supervision.

- Ethical and safety concerns are evident, as the majority advocate for the use of AI as support, conditioning trust in exclusively automated developments on rigorous validation. Justification of the sample

16.3 Justification of the Sample Size

1. Determination of the Population: Infinite vs. Finite

The target population of the study includes medical professionals, researchers in

biomedicine, and people related to the application of AI in medicine. Since it is not

limited to a closed group (e.g., employees of a specific hospital), it is assumed

an infinite population or sufficiently large to apply standard sampling formulas

without adjustments for finiteness.

2. Calculation of Theoretical Sample Size

To estimate the minimum required sample size, the formula for

infinite populations is used:

$n = (z^2 \times p \times (1 - p))/e^2$

- Confidence level (z): 95%, thus $z = 1.96$.

- Estimated proportion (p): 0.5 (maximum variability, as there are no prior data).

- Margin of error (e): 10% (0.10), acceptable in exploratory studies. Then, $n = ((1.96)^2 \cdot 0.5 \cdot 0.5)/(0.10)^2 \approx 96.04$.

Interpretation:

- Theoretically, 96 respondents would be needed to guarantee a 10% margin of error with 95% confidence.

3. Obtained Sample: 76 Respondents

The collected sample (n=76) is below the theoretical calculation, which could be attributed to:

- Practical limitations: Difficulty accessing more participants within the project timeframe.

- Study focus: Exploratory and descriptive, where absolute precision is not critical.

Sample Validity:

Although below ideal, n=76 is still adequate for:

- Descriptive analyses: Proportions, percentages, and general trends.

- Pattern identification: Basic correlations and thematic groupings.

- Triangulation with qualitative data: Expert interviews complement quantitative findings.

4. Expert Interviews (n=3)

In qualitative research, sample size is guided by the principle of saturation (when

new data provide no novel information). For this study:

- Intentional selection: The 3 experts have specialized profiles in biomedicine, guaranteeing relevance.

- Depth over quantity: Their answers provide detailed insights on technical, ethical, and practical challenges, enriching the quantitative analysis.

- Contextual validity: Their experiences reflect realities of the field, validating the applicability of the results.

5. Consideration of Variability and Margin of Error

- Variability (p=0.5): By using the maximum value (p=50%), the worst possible variability is assumed, which justifies a broader margin of error (10%).

6. Conclusion

The sample of 76 respondents and 3 experts is valid for the objectives of the study because:

1. It represents general trends in perceptions about AI in medicine, with an acceptable margin of error (10%) for exploratory purposes.

2. It triangulates methods: It combines quantitative analysis (surveys) and qualitative (interviews), compensating for size limitations.

3. It aligns with available resources: It respects practical constraints without compromising academic rigor.

Interviews

16.4 Interview 1

I conducted an interview with a laboratory technician friend, who currently works at the Hospital of Móstoles. His name is Daniel Labella.

Figure 15: Daniel Labella

Before we start, do you authorize the use of your name in the results of this research?

Sure, I authorize the use of my name in the results.

Perfect. Do you agree to participate in this interview, understanding that your answers will be used exclusively for scientific research purposes?

Yes, I completely agree to participate.

16.4.1 Introduction: Knowledge about AI in Medicine

To begin, could you tell me if you are familiar with the use of artificial intelligence in medicine?

Yes, I am quite familiar, especially with the use of AI in medical image analysis.

And in your opinion, what are the areas of medicine that are currently benefiting the most from the use of AI?

Well, nowadays we can see significant transformations in various areas. For example, AI is revolutionizing imaging diagnosis, and it is also being used in cardiology, oncology, neurology, and in such innovative fields as precision medicine and genomics. It is also being applied in primary care and telemedicine, in dermatology, psychiatry, robot-assisted surgeries, and even in epidemiology and public health.

Interesting. Have you had direct experience or heard about the application of AI in molecular structure analysis or drug development?

No, I haven't actually worked directly in that field, but I am using AI mainly in medical image diagnosis.

16.4.2 Part 1: Impact of AI on Biomedical Research

Moving on to biomedical research, how important do you think AI is for accelerating processes such as the design of new drugs?

AI is completely changing the landscape in drug design. It is accelerating

processes that used to be long, costly, and with high failure rates. Among its main

advantages are the reduction of time and costs, increased accuracy, innovation in

the search for new drugs, and, of course, its impact on personalized medicine. However,

there are also challenges and considerations that still need to be addressed.

I understand. According to your experience, what limitations have you observed when employing machine learning models in biomedical tasks?

I have noticed several important limitations. On the one hand, the quality and availability of data is

crucial, and sometimes it is not ideal. There are also problems with interpretability and

transparency of some models. Additionally, there are limitations in generalization, ethical

and legal issues, and difficulties in validation and clinical acceptance. We cannot forget the

technical limitations and integration into clinical workflows, as well as the challenges

that arise during clinical trials.

I see. How do you see the balance between accuracy and efficiency in AI tools in

this field?

The balance is fundamental and depends heavily on the purpose and context. The most

effective solutions are those that manage to maximize both accuracy and efficiency within the

constraints of the system in which they are implemented. The essential thing is to align the

model's capabilities with the practical needs of the clinical environment.

16.4.3 Part 2: Value of the Proposed Spiking Neural Network Model

Now, considering that Spiking Neural Networks can reduce by half both the

processing time and resource consumption compared to Transformers,

what impact do you think this would have on biomedical research?

The adoption of Spiking Neural Networks could be truly revolutionary. Their

efficiency in terms of time and resources would have several positive impacts: it would accelerate the

analysis of biomedical data, reduce operational costs, and democratize the use of AI in

biomedicine. Furthermore, it would drive specific applications, enable scalability in

large-scale research, and perform better in low-latency environments. All of this opens

the door to important future implications in research and development.

Very interesting. Do you think that features such as molecular weight, TPSA, logP, and the

number of atoms are sufficient to predict values like pIC50? Would you add other

variables?

Although those features are useful and show good correlation with biological activity,

in my opinion they are not sufficient to capture all the complexity of the molecular

interactions necessary to accurately predict pIC50. It is likely that other

complementary variables would be required to obtain a more accurate prediction.

Finally, how important do you consider speed and resource consumption in the

adoption of new models in research?

They are extremely important, especially in biomedicine, where resources are often

limited and every minute counts to achieve significant advances or even save lives.

16.4.4 Part 3: Ethics and Trust in AI in Medicine

Moving on to a crucial topic, do you think it is ethical to rely on AI-based tools

to make critical decisions in drug development?

Yes, I consider it ethical to use AI for making critical decisions, as long as it is ensured that

the tools are properly validated and supervised by experts.

Would you trust a drug developed entirely by AI?

No, fully trusting a drug developed exclusively by AI represents an

ethical, technical, and regulatory challenge. Personally, I would not trust it without rigorous human and

experimental validation.

And, do you think clear limits should be established regarding the role of AI in processes that

directly affect people's lives?

Absolutely. It is fundamental to establish clear limits to guarantee a responsible and

ethical use of AI in medicine, especially in decisions that directly impact the lives

of people.

16.4.5 Additional Questions

To conclude, what would you say are the main challenges to integrating more

efficient models into the development of new drugs?

I believe the main challenges include the quality and quantity of the available data, the

need for explainability and trust in the models, integration into existing workflows,

experimental and regulatory validation, and issues related to

scalability and sustainability. Cultural, ethical acceptance and, of course, competition and funding

are also important.

And what measures do you consider necessary to ensure that the results generated by AI

are reliable and ethically acceptable?

To achieve this, it is crucial to combine transparency, human supervision, strict regulations, and

strong ethical principles. Only in this way can the reliability and ethical acceptance of the

results be guaranteed.

If you had to choose between a faster but less accurate model and a more accurate but

slower one, which would you prefer?

In the field of biomedicine, accuracy is fundamental. That is why I would prioritize a more

accurate model, since safety and efficacy are much more important than simply reducing

processing times.

Thank you very much for your time and for sharing your valuable insights.

Thank you. I hope my answers are useful for your research.

16.5 Interview 2

Two professionals, coworkers of the one from the previous interview, have been interviewed; however, they

have not wanted to make their names public. They are equally reputable and medical students.

16.5.1 Authorization and Participation

Interviewer:

Before we begin, do you authorize the use of your name in the results of this research?

- Expert 1: Yes, I authorize the use of my name, but I prefer to keep certain personal details private.
- Expert 2: Sure, I have no problem with my name being used for the research.

Interviewer:

Do you agree to participate in this interview, understanding that your responses will be used

exclusively for scientific research purposes?

- Expert 1: Yes, I agree and I think it is important to share our experience in this field.
- Expert 2: Of course, I consider it essential to contribute ideas and experiences in the field of AI applied to medicine.

16.5.2 Introduction: Knowledge about AI in Medicine

Interviewer:

Are you familiar with the use of artificial intelligence in medicine?

- Expert 1: Yes, I am familiar, especially in the area of image diagnosis; however, I still feel a bit reluctant to use it in areas that involve critical decisions without sufficient support.
- Expert 2: Yes, I have quite a bit of experience. I regularly use AI to analyze medical images and support its integration into other clinical processes.

Interviewer:

Which areas of medicine do you think are benefiting the most from the use of AI currently?

- Expert 1: I believe that image diagnosis and cardiology have benefited remarkably. I also see potential in oncology and neurology, but still with caution in their full application.
- Expert 2:

In my opinion, AI is revolutionizing image analysis, diagnosis, and drug design. Moreover, it extends to areas such as precision medicine, telemedicine, and public health, where a positive impact is already noticeable.

Interviewer:

Have you worked or heard about the implementation of AI for the analysis of molecular structures or drug development?

- Expert 1: I have not worked directly in that area, although I know the studies and see that it is a field with much potential. I prefer to stay in applications where I already have experience, such as diagnosis.
- Expert 2: Yes, I have closely followed projects that use AI to analyze molecular structures and optimize drug design, and I am enthusiastic about the idea of integrating these techniques into my practice.

16.5.3 Part 1: Impact of AI on Biomedical Research

Interviewer:

How important do you think AI is for accelerating processes such as the design of new drugs?

- Expert 1: AI has great potential to reduce times and costs, but I am concerned that, without proper supervision, critical details could be overlooked. It is an important advance, although it must be implemented with caution.
- Expert 2: I consider that AI is transformative in drug development. I already see it in action, helping to drastically reduce times and costs, and providing greater precision in the identification of therapeutic candidates.

Interviewer:

According to your experience, what limitations have you observed when using machine learning models in biomedical tasks?

- Expert 1: I have noticed that the quality and availability of data is a key limitation, along with the difficulty in interpreting certain models. Furthermore, clinical validation is still a major challenge.
- Expert 2: The limitations are mainly in data quality and integration with clinical systems. Although interpretability is improving, I consider that the technology continues to evolve rapidly to overcome these obstacles.

Interviewer:

How do you perceive the balance between accuracy and efficiency in AI-based tools in this field?

- Expert 1: It is crucial to find a balance; in my experience, accuracy should not be sacrificed for speed, especially in critical decisions. Efficiency is important, but it should always be accompanied by rigorous validation protocols.
- Expert 2: Personally, I believe it is possible to achieve both without compromising safety. I prioritize precise models, but I have also seen that they can be optimized to be faster without losing reliability.

16.5.4 Part 2: Value of the Proposal of the Spiking Neural Network Model

Interviewer:

Knowing that Spiking Neural Networks reduce time and resource consumption compared to

Transformers, what impact do you think this could have on biomedical research?

- Expert 1: I think it is an interesting proposal that could improve efficiency, but I have reservations about whether its application can be so broad in clinical environments without compromising safety.

- Expert 2: I am very excited about this technology. Reducing resources and time could revolutionize biomedical data analysis and allow larger-scale studies to be conducted at lower costs.

Interviewer: Do you believe that features such as 'mol_weight', 'TPSA', 'logP' and 'num_atoms' are sufficient to predict
values like pIC50? Would you add others?

- Expert 1: They are useful indicators, but not sufficient to capture all the complexity of molecular interactions. It is probably necessary to include additional variables, such as three-dimensional conformation or specific interactions between molecules.

- Expert 2: In my opinion, those features are a good foundation, but they could definitely be complemented with molecular dynamics data and biophysical parameters to improve the accuracy in predicting pIC50.

Interviewer:
How important are speed and resource consumption to you when adopting new
models in research?

- Expert 1: They are very important, especially in a clinical setting where every minute counts. However, accuracy should always be prioritized.

- Expert 2: Absolutely crucial. In biomedicine, where resources can be limited, having efficient and fast models makes a difference, without compromising the accuracy of the results.

16.5.5 Part 3: Ethics and Trust in AI in Medicine

Interviewer:
Do you consider it ethical to rely on AI-based tools to make critical decisions in drug
development?

- Expert 1: Yes, as long as they are implemented with rigorous supervision and validation. It is ethical, but we must be very careful with the protocols.

- Expert 2: Totally agree. Ethics is not questioned if AI is used as support and with human validation. I believe it can be a valuable tool if applied correctly.

Interviewer:
Would you trust a drug entirely developed by AI?

- Expert 1: No, I would only trust a drug developed by AI if it has undergone rigorous testing and has the necessary human validation.

- Expert 2: Although the technology is promising, I personally would not blindly trust a drug developed exclusively by AI without extensive verification and clinical supervision.

Interviewer:
Do you think clear limits should be established regarding the role of AI in processes that directly affect
human lives?

- Expert 1: Definitely. It is fundamental to establish limits to avoid risks and ensure that AI is used responsibly in medicine.

- Expert 2: Yes, establishing clear limits is crucial to ensure that AI complements human work and does not replace clinical judgment in critical situations.

16.5.6 Additional Questions

Interviewer:
What do you believe are the main challenges to integrating more efficient models in the development of
new drugs?

- Expert 1: The main challenges are the quality and quantity of data, the lack of explainability in some models, and integrating these systems into clinical workflows without disrupting medical care.

- Expert 2: I think the biggest challenges are experimental validation, scalability, and the need for funding and ethical acceptance. Undoubtedly, they are challenges but also opportunities to innovate.

Interviewer:
What measures do you consider necessary to ensure that results generated by AI are reliable and ethically acceptable?

- Expert 1: It is essential to combine transparency, human supervision, and strict regulations. Ethics must be integrated into every stage of model development.
- Expert 2: I agree; clear standards, regular audits, and continuous oversight are required to ensure reliability and ethics in the use of AI.

Interviewer:
Finally, if you had to choose between a faster but less accurate model and a more accurate but slower one, which would you choose?

- Expert 1: In my case, I would prioritize accuracy. In biomedicine, it is essential to guarantee safety and efficacy, even if that means longer processing times.
- Expert 2: I would choose the more accurate model, since accuracy is essential for clinical decision-making, even if it means waiting a bit longer.

Interviewer:
Thank you both very much for your time and for sharing such valuable perspectives.

- Expert 1: Thanks to you, it has been a pleasure to share my experience.
- Expert 2: Thank you very much, I hope our answers contribute to advancing the research.

17 Stakeholder Analysis and Management

The proper identification and management of stakeholders is vital for the success of any project. Not only does it allow us to know who the interested parties are and what their level of influence is, but it also helps us to draw up communication strategies that ensure effective and timely collaboration.

1. University / Academic Institution (e.g.: Carlos III University)

- Power: High The university is the institutional engine that supports the research, providing resources, infrastructure, and academic prestige. Its influence is crucial for the viability and recognition of the project.
- Interest: High The research results directly impact the image and reputation of the institution, opening doors to future collaborations and funding opportunities.
- Position: Proactive and collaborative The university is expected to be actively involved, contributing ideas and supporting decision-making, which fosters a team and innovation work environment.
- Mode and Frequency of Communication:
- Mode: Use formal meetings, written reports, and updates through institutional email.
- Frequency: Organize quarterly meetings and send monthly progress reports so all academic leaders are aware of the progress.
- Rationale: This approach allows the institution to regularly supervise the project, solve problems on time, and adjust the strategy when necessary. Transparency and constant communication strengthen trust and foster a collaborative environment that benefits all involved.

2. Pharmaceutical Company (e.g.: LabFarm Innovadores)

- Power: High Pharmaceutical sector companies have the capacity to transform scientific advances into practical and commercial applications, which can take the research to a higher level.
- Interest: High Innovation in drug-target interaction prediction is key to optimizing the development process of new drugs, which can mean significant savings in time and costs.

- Position: Strategic and results-oriented These companies seek solutions that translate into competitive advantages, so they expect to see concrete and viable results that can be applied in their R&D processes.

- Mode and Frequency of Communication:

- Mode: It is recommended to conduct executive presentations, technical workshops, and analysis sessions with interactive tools that facilitate understanding of the results.

- Frequency: Establish monthly meetings to review progress, complemented with quarterly workshops to delve into specific aspects and define joint strategies.

- Rationale: Maintaining constant and specialized communication with the company allows adjusting the research to real market needs. This helps identify opportunities and solve possible barriers early, ensuring that the developments are both technically sound and commercially viable.

3. Main Funder (e.g.: National Research Agency)

- Power: High Funders are the ones who make the project feasible, as they provide the necessary economic resources to carry it out.

- Interest: High The proper use of funds and obtaining innovative results are fundamental to justify the investment. Funders seek to see advances that bring value both to science and to society.

- Position: Demanding but favorable Although they require meticulous accountability, their stance is positive when promising results are demonstrated and established objectives are met.

- Mode and Frequency of Communication:

- Mode: Detailed progress reports must be sent and presentations of key milestones made in follow-up meetings, addressing both achievements and challenges.

- Frequency: Quarterly reports, along with specific meetings at the close of important project phases.

- Rationale: Clear and regular communication with the funder is essential to maintain their trust and ensure the continuity of financial support. This communication channel allows adjusting the strategy based on project performance and transparently demonstrating the proper use of resources.

4. Academic Community (Researchers, Professors, and Students)

- Power: Medium Although they do not directly influence project implementation, validation and constructive criticism from the academic community are essential for the recognition and dissemination of the research.

- Interest: High Members of the academic community are interested in knowledge generation, methodological innovation, and the possibility of establishing collaborations that enrich the research work.

- Position: Constructive and collaborative criticism They are expected to offer well-founded feedback and valuable suggestions, participating in debates that help improve the quality and scope of the research.

- Mode and Frequency of Communication:

- Mode: Organize seminars, conferences, and round tables, in addition to publishing articles in specialized journals.

- Frequency: Semiannual updates through academic events and periodic publications that allow constant exchange of ideas.

- Rationale: Fostering an open and continuous dialogue with the academic community enriches the project, allows hypotheses to be validated, and strengthens the collaboration network. Regular interaction creates an atmosphere of trust and stimulates creativity, which benefits both scientific quality and the training of future researchers.

5. General Public and Patients

- Power: Low to Medium Although they do not have a direct impact on technical decisions, the perception and acceptance of the general public are fundamental for the legitimacy and long-term success of scientific advances.

- Interest: Medium This group is interested in how research can improve quality of life and offer new solutions to health problems. However, they require information to be presented clearly and accessibly.

- Position: Cautious but optimistic The general public and patients tend to be skeptical of innovations that are not communicated transparently but are optimistic when they understand the potential benefits.

- Mode and Frequency of Communication:

- Mode: Use press releases, social media posts, informative talks, and outreach events that facilitate understanding of complex concepts through infographics and explanatory videos.

- Frequency: Conduct quarterly communication campaigns and organize annual events where progress is explained and questions answered interactively.

- Rationale: An effective communication strategy with the general public is crucial to generate trust in science and promote the acceptance of new technologies. Educating the community and patients about the benefits and the potential impact of the research fosters a positive perception and the social support necessary for the implementation of innovations in health.

18 Evaluation of Results and Future Improvements

(Based on the results obtained in the attached Google Colab)

I have verified that Spiking Neural Networks (SNN) not only outperform the Transformer model in

terms of computational efficiency, but also maintain high accuracy in predicting

drug-target interactions. This translates into significantly shorter training times

and reduced memory usage, essential aspects for applications in resource-limited environments.

Highlighted Aspects:

- Resource Efficiency: SNNs use up to 50% less RAM compared to the Transformer, which is crucial for implementation in systems with hardware limitations.

- Reduced Training Time: The SNN model trains in less than half the time required by the Transformer, allowing for faster iterations and greater agility in the development process.

- High Level of Accuracy: Despite the remarkable efficiency, the SNN demonstrates comparable and even superior accuracy in certain cases, ensuring its effectiveness for applications in drug discovery.

Future Improvements:

- Hyperparameter Optimization: Perform fine-tuning of the SNN model parameters to maximize predictive capability and stability across different scenarios, which could further improve its performance.

- Expansion and Diversification of the Dataset: Increase the variety and number of molecules and SMILES in the analysis to strengthen the model's generalization ability and validate its application in broader contexts.

- Testing in Real-World Environments: Implement pilot studies in clinical settings or pharmaceutical development to evaluate model performance under real-world conditions, which will help identify new challenges and adjust the methodology practically.

- Exploration of Specialized Hardware: Investigate the integration of neuromorphic hardware or other

emerging technologies that further optimize energy consumption and model performance, making the solution more scalable and accessible.

18.1 Opportunities and Comparison with Results

The findings obtained in the Google Colab confirm that SNNs not only meet the computational efficiency goals, but also exceed the expectations established at the

beginning of the project. This opens the following opportunities:

1. Deployment in Resource-Limited Environments

The reduction of up to 50% in RAM usage implies that SNNs can be integrated

into devices with modest hardware (e.g., portable diagnostic stations or laboratories

with limited computational capacity), democratizing access to advanced pharmacological

prediction tools.

2. More Agile Development Cycles

Training the SNN in less than half the time required for the Transformer facilitates

rapid iterations during experimentation and model tuning. This agility allows

for faster responses to new hypotheses and accelerates validation of compounds in early

stages of drug design. Or even accelerating new drugs in case of

need (e.g., Covid 2019).

3. Maintenance of High Accuracy

Despite its efficiency, SNNs achieve a lower val_loss than the Transformer, indicating

a superior level of accuracy. This creates the opportunity to rely on SNNs both for exploratory tasks and for applications where accuracy is critical, such as predicting side effects or selecting promising candidates.

4. Innovation in Hardware and Methodologies

Given the low energy consumption of SNNs, there is an opportunity to explore neuromorphic hardware and other emerging architectures that further enhance this advantage. Likewise,

the door opens to hybrid methodologies combining SNNs with Transformers at different stages of the workflow (e.g., SNN for rapid screening and Transformer for fine analysis).

Opportunity Obtained Result

Deployment in resource-

limited environments

SNN consumes less RAM (4.60 MB vs. 464.55 MB), demonstrating

feasibility on modest hardware.

More agile development cycles SNN training time at 29.14 s, less than half that of

the Transformer (60.62 s).

Maintenance of high accuracy Val_loss of SNN (1.8311) lower than Transformer (5.2545),

ensuring accuracy in predictions.

Innovation in hardware and

methodologies

Low SNN consumption opens the door to use of neuromorphic hardwareand hybrid AI schemes.

19 Reflection

19.1 Reflection on the research process

Looking back, the development of this project has been a formative experience both

academically and personally. The process did not follow a straight line; it was a constant spiral of

planning, execution, review, and learning. From designing the research question to the

analysis of the results, there were moments when I was forced to adjust the course. For example,

during the Transformer model training phase, I observed that its resource consumption greatly exceeded

expectations. This situation led me to redirect the analytical focus towards the SNNs, not as

a simple alternative, but as a realistic and efficient solution. This was a clear instance of reflection

in action, where I made technical decisions in real time guided by intuition and contextual knowledge.

In turn, when analyzing the final results, I practiced reflection on action. I stopped to consider

not only what had worked, but also which factors influenced those results and how I could

improve in future projects. One of the most valuable lessons was understanding that research does not

end when results are obtained, but rather these mark the beginning of new questions, challenges, and

directions. In that sense, I now understand what authors like McNiff point out: our findings

must remain "lightly held," ready to be questioned and updated.

19.2 Reflection on the researcher's performance

Regarding my role as a researcher, I have learned to combine technical rigor with the flexibility

needed to adapt to the unexpected. At the beginning of the project I underestimated the complexity of

implementing models like the SNNs, both in data preparation and in the need to adjust

hyperparameters. This forced me to develop greater technical autonomy, but also to seek

support in literature and external resources. Time management was a constant challenge, especially when

balancing coding tasks, qualitative analysis, and academic writing.

I realize I still need to improve key skills such as long-term planning, more

strategic use of scientific bibliography, and the application of more advanced statistical techniques. However,

I feel satisfied with the level of depth achieved, both in technical analysis and in the

ethical and social interpretation of the results. I was able to apply a professional, but also human,

approach, always seeking that my findings were not only valid, but also useful and understandable for

different audiences.

This project leaves me with new questions: How would this model behave with real

clinical datasets? How does its technical performance translate into therapeutic impact? Could SNNs collaborate

with other architectures to design even more powerful solutions? These and other future lines invite me

to continue researching, knowing that each answer is only the start of a new exploration.

20 Future Lines of Improvement

1. Expansion and diversification of the dataset

To improve SNN generalization even further, it is recommended to integrate new repositories (ChEMBL,

PubChem, ZINC) and enrich the initial dataset with advanced descriptors. Likewise, it is advisable to balance

the distribution of pIC_{50} values using oversampling techniques (SMOTE) and generate chemically validated synthetic SMILES.

All this should be implemented in a reproducible pipeline (for

example, in Python/RDKit) that documents every step of cleaning, normalization, and quality verification.

2. Validation in real environments and pilot testing

It is essential to bring the SNN predictions to an R&D laboratory to evaluate their accuracy in

real biochemical assays (binding assays, IC_{50} tests). This will allow quantitatively comparing the

predictions with experimental results and establishing clear success thresholds (e.g., mean absolute deviation ¡10 %).

Additionally, incorporating the SNN into the workflow of a pharmaceutical area will allow

measuring response times and collecting direct feedback from chemists and biologists, adjusting the

model with practical data.

3. Exploration of specialized hardware and energy efficiency

To leverage the inherent efficiency of SNNs, implementation should be migrated to neuromorphic chips (IBM, 2025) (Intel Loihi, IBM TrueNorth) and their energy consumption (Wh/molecule) compared comparatively against conventional GPU environments. Similarly, it is advisable to port

quantized and pruned versions of the SNN to other smaller devices (Raspberry Pi, Jetson Nano),

measuring inference times and memory under real portable laboratory conditions. This analysis

will validate the viability of using SNNs in low-consumption environments.

4. Strengthening interdisciplinary collaboration

It is essential to organize practical workshops in which chemists, computational biologists, and AI engineers

collaborate to refine molecular descriptors, interpret peak patterns, and define success criteria. Creating a network of external validators (pharmacologists and ethics experts) who reviewquarterly predictions of the SNN will guarantee clinical and ethical rigor. Likewise, the continuous publication of open data and partial results will promote feedback from the scientific community, enriching the project with multiple perspectives.

5. Tests in clinical stress environments and pandemic drills

Design test scenarios that emulate high-pressure situations, such as an epidemic outbreak, where the speed of response and the accuracy of predictions are critical. This involves simulating massive peaks of biological data (such as viral sequences or patient data) to verify that the SNN maintains low val_loss and minimal inference times, validating its potential in emergencies similar to COVID-19. With these drills, the model can be fine-tuned for environments where every second and every precise prediction can make a difference.

21 Evaluative Conclusion

This research has made it possible to establish, in an objective and well-founded manner, that Spiking Neural Networks present important advantages over Transformers in the prediction of interactions between drugs and biological targets. Quantitative analyses show a significant reduction in training time and memory usage, while qualitative results confirm that the accuracy of the SNN is at least comparable and sometimes superior. These findings are based on solid evidence and rigorous criteria that have allowed assessing both the efficacy and the quality of the investigated proposal.

It is relevant to recognize that, despite their strengths, the implementation of SNNs entails challenges in terms of technical complexity and scalability. The critical evaluation carried out highlights the need to continue refining these models to ensure their integration into real biomedical environments. In summary, this project not only demonstrates the potential of SNNs to transform the drug discovery process but also provides a clear basis for future research that can optimize their application and expand their benefits in the field of artificial intelligence applied to biomedicine.

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